

Are Apple's Quarterly Returns Forecastable?

A Time-Series Investigation, 1981–2026

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STAT 248 — Spring 2026

May 13, 2026

The research question

Question

Do Apple's quarterly stock returns contain stable temporal dependence and seasonality that are strong enough to produce out-of-sample forecasts that outperform simple naive benchmarks on recent quarters?

What a “yes” would look like: an ARIMA or regularized regression beats both the historical-mean and persistence baselines on MAE, RMSE, and a Diebold–Mariano test, with white-noise residuals.

What a “no” would look like: models tie or trail the naive benchmarks; gains are not stable across nearby test windows; residuals still carry structure.

Why Apple, why quarterly?

- **Single firm, not an index.** Avoids cross-sectional aggregation; isolates one company's earnings cycle.
- **Quarterly cadence.** Aligns with Apple's reporting calendar and with product-launch patterns (iPhone in Q4, education in Q3).
- **Long sample.** 1981Q1–2026Q2 spans the entire public history of Apple as a listed firm.
- **Tractable size.** 182 observations is enough for ARIMA but small enough to expose the limits of complex models.

Why a time-series problem?

- Returns are *ordered* in time; OLS independence assumption does not hold by default.
- Seasonal effects (fiscal-quarter dummies) and lagged returns are natural candidate features.
- Forecasting is a *strictly causal* task: at quarter t , only information available before t may be used.
- Ignoring temporal dependence gives biased standard errors and optimistic out-of-sample claims.

Talk roadmap

- ① **Data** — the Apple quarterly panel and the target variable.
- ② **EDA & diagnostics** — stationarity, autocorrelation, seasonality, conditional heteroskedasticity.
- ③ **Models** — naive benchmarks, ARIMA / SARIMA, lagged regressions with regularization.
- ④ **Evaluation** — holdout, rolling-origin backtest, Diebold–Mariano test.
- ⑤ **Conclusion** — answer, limitations, future work.

Source. Kaggle “Apple Financial Dataset (1980–2026)”.

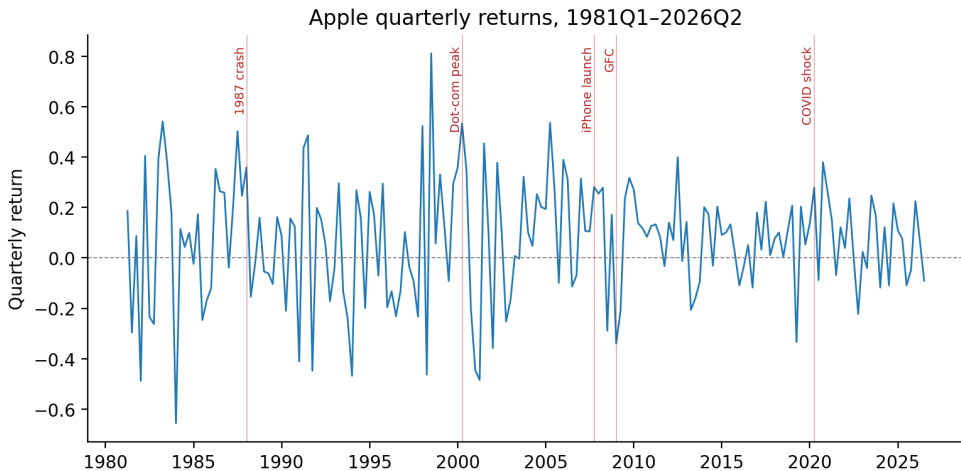
- `aapl_quarterly_master.csv` — primary table, **182 quarters**, 1981Q1 – 2026Q2.
- `aapl_quarterly_summary.csv` — product revenue shares (only complete from 2009Q1; used for sensitivity).
- Daily-level files used only for context.
- No missing values in the target.
- Quarter index built from (`fiscal_year`, `fiscal_quarter`) pairs; no duplicates, strictly monotonic.

The target variable: `quarter_return`

Definition. `quarter_return` is Apple's compounded return over each fiscal quarter, computed from intra-quarter daily returns.

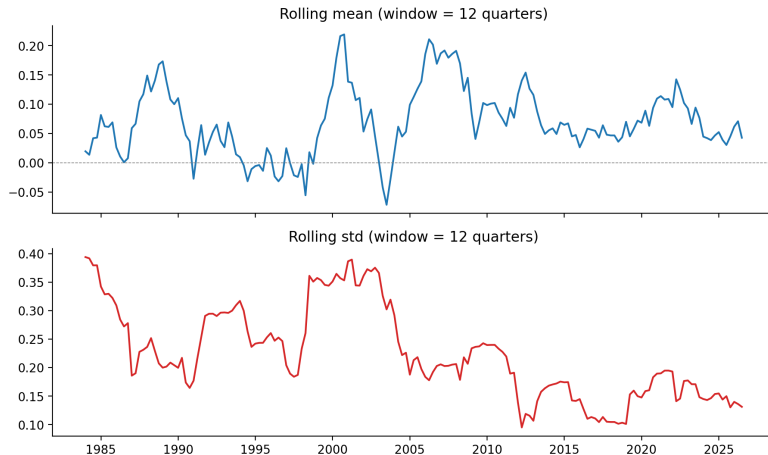
Statistic	Value
n	182
Mean	0.067
Std. dev.	0.236
Min	-0.656
Max	+0.812

45 years of quarterly returns



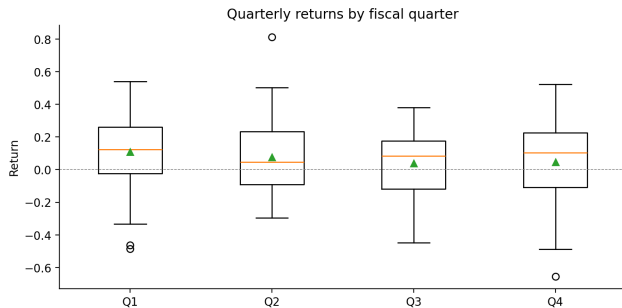
Vertical lines: 1987 crash, dot-com peak, GFC, COVID shock, iPhone launch.

Rolling mean and rolling volatility



12-quarter rolling mean is usually above zero; both rolling statistics are not flat.

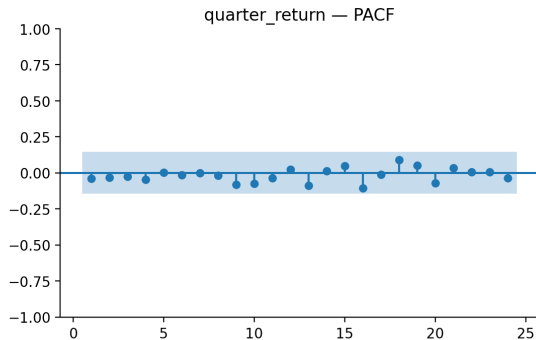
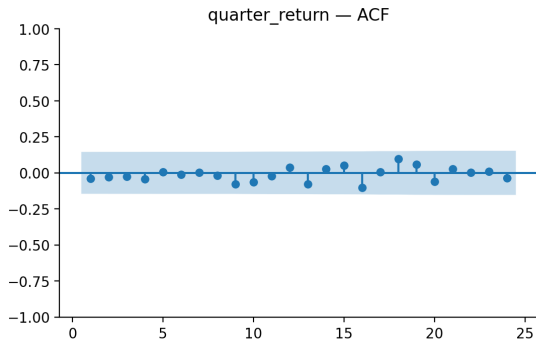
Seasonal pattern by fiscal quarter



Quarter	Mean	Std
Q1	0.108	0.236
Q2	0.076	0.237
Q3	0.038	0.197
Q4	0.046	0.270

Means differ across quarters but within-quarter variability is roughly 5× the between-quarter spread. Seasonality is real but weak.

ACF and PACF



No spike pierces the 95% confidence band at any lag up to 24. Sample ACF at lag 1 is -0.04 .

Stationarity tests agree

Test	Statistic	p -value	Conclusion
ADF (H_0 : unit root)	-13.90	$< 10^{-25}$	stationary
KPSS (H_0 : stationary)	0.09	0.10	stationary

Both tests agree. No differencing needed for the level of returns. Models can be fitted directly to `quarter_return`.

Mean is white noise; variance is not

Diagnostic	p -value	Reject H_0 ?
Ljung–Box (lag 4)	0.93	no
Ljung–Box (lag 8)	0.998	no
Ljung–Box (lag 12)	0.99	no
Engle ARCH (lag 12)	0.04	yes

Implication. Whatever predictability exists lives in the *second moment*, not the first.

Train / test split and backtest design

- **Holdout split.** Train: 1981Q1 – 2022Q4 ($n = 168$). Test: 2023Q1 – 2026Q2 ($n = 14$).
- **Rolling-origin backtest.** Expanding window starting at 2018Q1, refit at every step, 34 one-step forecasts.
- All hyperparameters tuned on *training* data only, with `TimeSeriesSplit` (no random shuffling).
- Two horizons let us check the result is not an artifact of one test window.

Three naive benchmarks

- **Historical mean.** $\hat{y}_t = \bar{y}_{\text{train}}$. Optimal under i.i.d. returns.
- **Persistence (naive last).** $\hat{y}_t = y_{t-1}$. Optimal under a random walk.
- **Seasonal naive.** $\hat{y}_t = y_{t-4}$. Optimal if returns are perfectly periodic with period 4.

These set the bar that any “real” model must clear.

ARIMA and SARIMA grid search

ARIMA grid: $p \in \{0, \dots, 3\}$, $d \in \{0, 1\}$, $q \in \{0, \dots, 3\}$ (32 specifications).

SARIMA grid: same plus seasonal $(P, D, Q) \in \{0, 1\}^3$ at $s = 4$ (256 specifications).

Selected model	AIC	BIC
ARIMA(1, 0, 1)	5.90	18.40
SARIMA(0,1,2) $\times$ (1,0,1) ₄	4.01	19.38

Both choose low-order specifications — consistent with the flat ACF.

Lagged regression family

Design matrix (per quarter t):

$$\mathbf{x}_t = (y_{t-1}, y_{t-2}, y_{t-3}, y_{t-4}, \mathbf{1}_{Q2}, \mathbf{1}_{Q3}, \mathbf{1}_{Q4})^\top$$

- **OLS** — baseline regression on the seven features.
- **Ridge** — ℓ_2 penalty, α grid 10^{-3} to 10^3 on a log scale.
- **LASSO** — ℓ_1 penalty, α grid 10^{-4} to 10^1 .
- Hyperparameters tuned with `TimeSeriesSplit` (5 expanding folds), *not* random K-fold.

Point-error metrics:

- MAE — mean absolute error
- RMSE — root mean squared error
- sMAPE — symmetric MAPE (robust near zero)

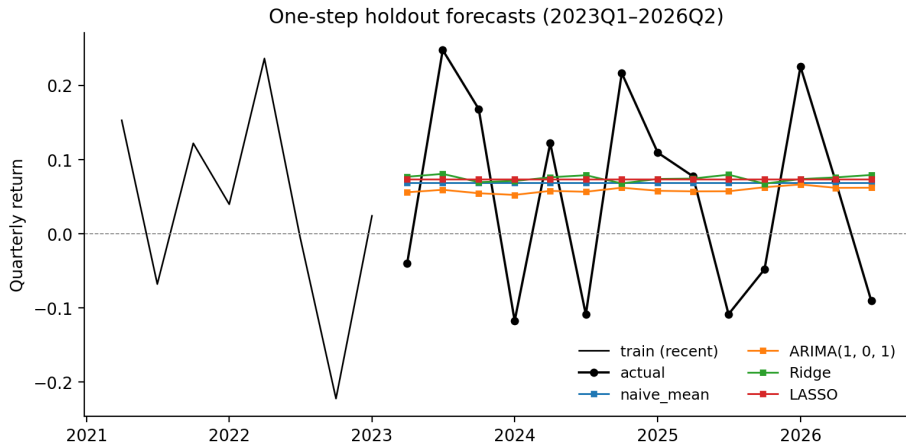
Significance test: Diebold–Mariano with the Harvey–Leybourne– Newbold (HLN) small-sample correction.

H_0 : equal predictive accuracy versus the historical-mean benchmark. Two-sided p -value reported.

Holdout metrics: 14 quarters from 2023Q1

Model	MAE	RMSE	RMSE vs. mean
naive_mean (baseline)	0.115	0.131	0.0%
naive_last	0.178	0.201	-53.2%
naive_seasonal	0.165	0.208	-58.4%
ARIMA(1,0,1)	0.116	0.129	+1.7%
SARIMA(0,1,2)(1,0,1) ₄	0.112	0.131	-0.0%
OLS	0.123	0.141	-7.1%
Ridge	0.116	0.133	-1.4%
LASSO	0.115	0.132	-0.5%

Forecasts on the holdout window



Black markers: realized quarterly returns. Colored lines: one-step-ahead forecasts. Every model hugs the historical-mean line.

Diebold–Mariano test on the holdout

Model vs. naive_mean	DM stat	HLN p
ARIMA(1,0,1)	−0.79	0.46
SARIMA(0,1,2)(1,0,1) ₄	0.00	0.998
OLS	1.20	0.27
Ridge	0.87	0.42
LASSO	0.56	0.60
naive_last	3.09	0.011
naive_seasonal	1.98	0.078

No “real” model significantly beats the historical mean. `naive_last` is significantly *worse*.

Robustness: rolling-origin backtest

- Expanding training window starting at 2018Q1.
- Refit every model at every quarter, predict one step ahead.
- 34 forecasts spanning pre-COVID, COVID-shock, and recent quarters.
- Goal: rule out the possibility that the holdout ranking is an artifact of a single test window.

Rolling-origin metrics: 34 quarters from 2018Q1

Model	MAE	RMSE	RMSE vs. mean
naive_mean (baseline)	0.126	0.155	0.0%
naive_last	0.198	0.237	-53.2%
naive_seasonal	0.179	0.228	-47.7%
ARIMA(1,0,1)	0.126	0.153	+0.7%
SARIMA(0,1,2)(1,0,1)₄	0.122	0.153	+ 0.8%
Ridge	0.127	0.156	-0.7%
LASSO	0.125	0.155	-0.0%

Same story: best model beats baseline by $\sim 1\%$, DM p around 0.7.

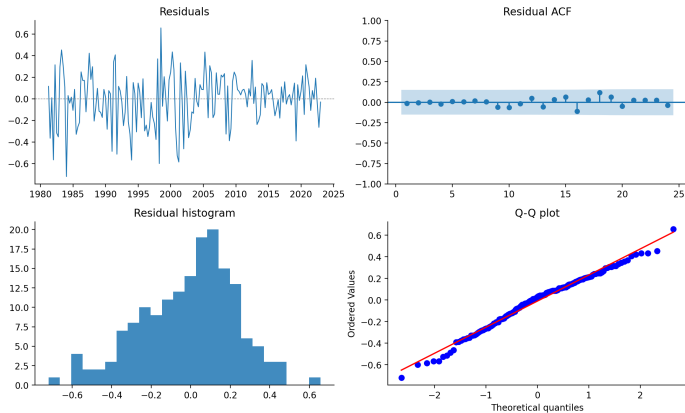
Why the regressions collapsed

Feature	OLS	Ridge ($\alpha = 10^3$)	LASSO ($\alpha = 0.083$)
lag 1	-0.007	-0.001	0.000
lag 2	-0.009	-0.002	0.000
lag 3	-0.004	-0.000	0.000
lag 4	-0.014	-0.001	0.000
Q2 dummy	-0.007	0.001	0.000
Q3 dummy	-0.035	-0.003	0.000
Q4 dummy	-0.024	-0.001	0.000

Ridge picks the largest α in the grid. LASSO drives every coefficient to exactly zero — it recovers the historical mean.

ARIMA residual diagnostics

ARIMA(1, 0, 1) residual diagnostics



Ljung–Box on residuals: $p > 0.99$ at all lags (white noise). **ARCH on residuals: $p = 0.009$ (still heteroskedastic).**

Answer to the research question

Verdict

No. Apple's quarterly returns over 1981–2026 do *not* contain temporal structure that translates into out-of-sample forecasts beating the historical mean.

Evidence that converges on the same answer:

- Flat ACF / PACF; Ljung–Box $p > 0.92$ at every tested lag.
- Best ARIMA improves RMSE by 1%–2%; DM $p \in [0.4, 0.7]$.
- Time-aware LASSO sets every coefficient to zero.
- Same conclusion on a single 14-quarter holdout and a 34-quarter rolling backtest.

Why this null result is informative

- Mean dynamics are statistically indistinguishable from white noise.
- Variance dynamics are demonstrably *not* white noise (ARCH $p < 0.05$ on raw and residuals).
- Predictability of Apple quarterly returns lives in the second moment, not the first.
- This is consistent with the efficient-market hypothesis on the mean and the volatility-clustering literature on the variance.

- **Sample size.** 168 training quarters is a fundamental ceiling on statistical power.
- **Structural break.** The 2020Q1 COVID shock dominates a handful of rolling-origin forecasts.
- **Missing covariates.** Product-revenue shares are available only from 2009Q1, so they cannot be used over the full sample.
- **Single ticker.** Conclusions are about Apple, not about equities in general.

Direct extensions:

- **GARCH(1,1)** on residuals — model the volatility structure we identified, evaluate value-at-risk forecasts.
- **State-space TVP models** that adapt to regime shifts (Markov-switching, time-varying parameters).
- **Earnings-window predictors** — analyst revisions and implied volatility ahead of each release.